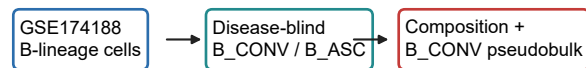


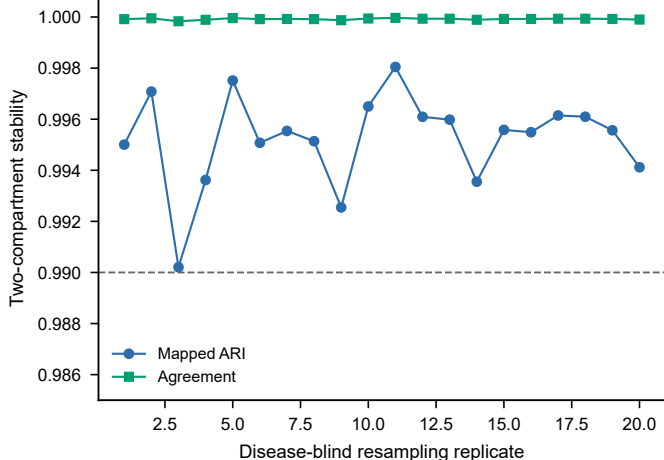
a Disease-blind discovery and frozen validation



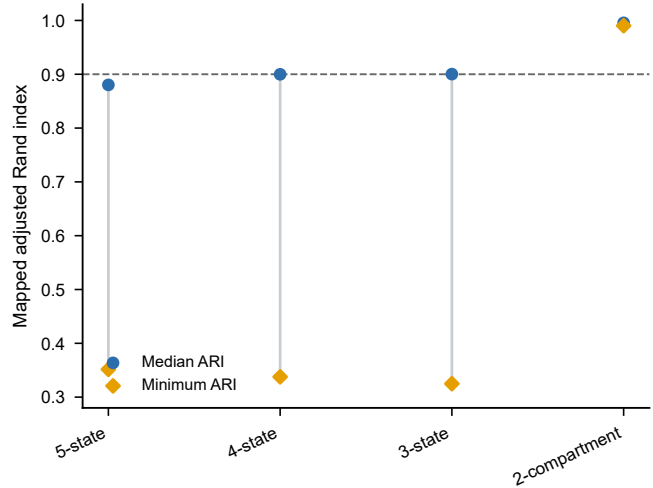
Frozen signatures → GSE135779 validation → regulatory convergence

259 donors | 271 samples | 332 sample-cohort strata | 88 libraries

c B_CONV/B_ASC reproduces in 20/20 runs



b Fine-state instability is retained



d Broad identities meet the frozen scope

